DS 3010 Homework 7 Solutions

Problem 1: Multiple Linear Regression

Suppose you work for a consulting firm. Your manager gives you the `Boston` dataset (part of library `ISLR2`) and asks you to build a multiple linear regression model to predict median home prices (`medv`). In typical form, he gives you no instructions or input. **The final deliverable should be a model that can accurately predict median home price from a set of predictors.**

Create a report (**no more than 1 page**) that includes the following items/discussion:

(a) Report your final model and its regression coefficients in a nicely summarized table.

(b) Not all predictors need to be in the model. What technique will you use to determine which predictors to include in the model and which to omit? Explain your workflow and justify how you decided on your "final" model. Directly address why you choose your final model over other candidate models.

(c) What assumptions were needed to obtain this final model? Are there any diagnostic checks you made to ensure those assumptions were valid? Did you update your final model to accommodate any assumption violations?

(d) Were there any potential issues related to your final model that you explored? Did you address these issues once identified?

(e) Can you obtain the bias and variance of this model? If so, explain how. If not, explain why not.

Since you are the only data scientist on the team, your manager reminds you to make every effort to produce a report that is rigorous but **understandable to non-statisticians**. Limitations/strengths of your pipeline to build this model should be discussed. Your workflow should be thoughtfully justified using the concepts/ideas we have discussed in class.

Solution: *There are many ways students could have gone about this problem. The main elements to look for are (1) do students understand the strength/limitations of their approach and (2) can they thoughtfully justify their workflow? Is their final model and its prediction accuracy well supported by the data? Limitations and potential issues should be discussed and considered. Below is my approach:*

The goal here is prediction. Therefore, my strategy is to build a model on a training set only and then evaluate its prediction accuracy on the test set. This will give me the most realistic evaluation of how well the model will perform on the prediction of new observations. Since the number of predictors (p) is less than 30, I will use subset selection to select a set of candidate models. Subset selection will directly obtain the 'best' predictors for each model size by finding the combination of predictors that minimizes the RSS (training MSE). Since subset selection is an exhaustive search, it will return to me the optimal model according to a specified criteria. I will use test MSE as my criteria to select my final model, since again our focus is on prediction. Following this approach led me to the model with the following predictors: ***`crim, zn, chas, nox, rm, dis, rad, tax,`***

*`ptratio, lstat`. This model had the smallest test MSE among all candidate models $M_1 - M_{12}$. Its test MSE was **22.00862**. See Figure 1 for a comparison of its test MSE with other models.*

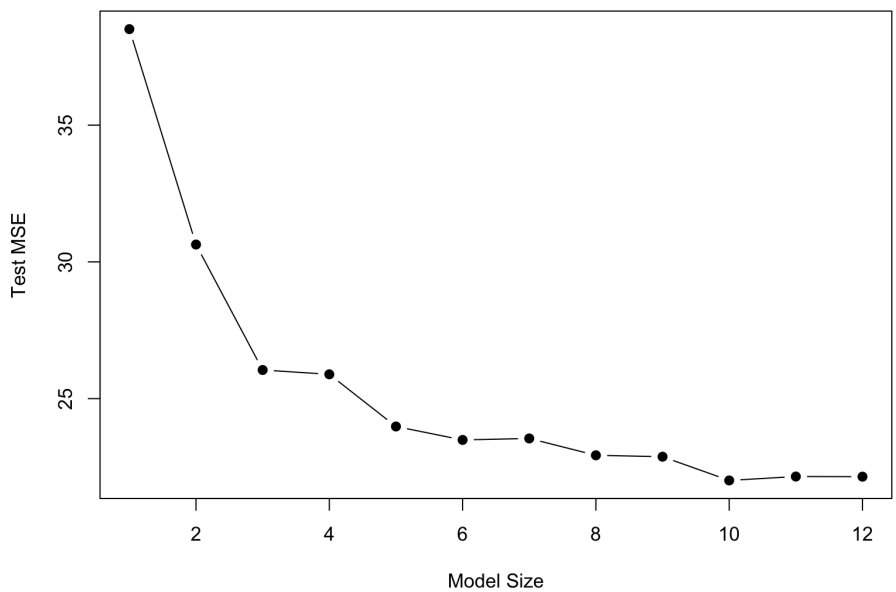

Figure 1: Test MSEs versus candidate models

Linearity and constant variance are assumed when fitting this model. A linear regression model on the above predictors (with medv as the response) was fit (on the full data) and diagnostics were examined. Please see Figure 2 (left). Clearly non-linearity is violated; non-constant variance may also be a concern. After some exploration and trial and error, I decided to add a polynomial term to `lstat`. The following diagnostic plots are a marked improvement (see Figure 2 right).

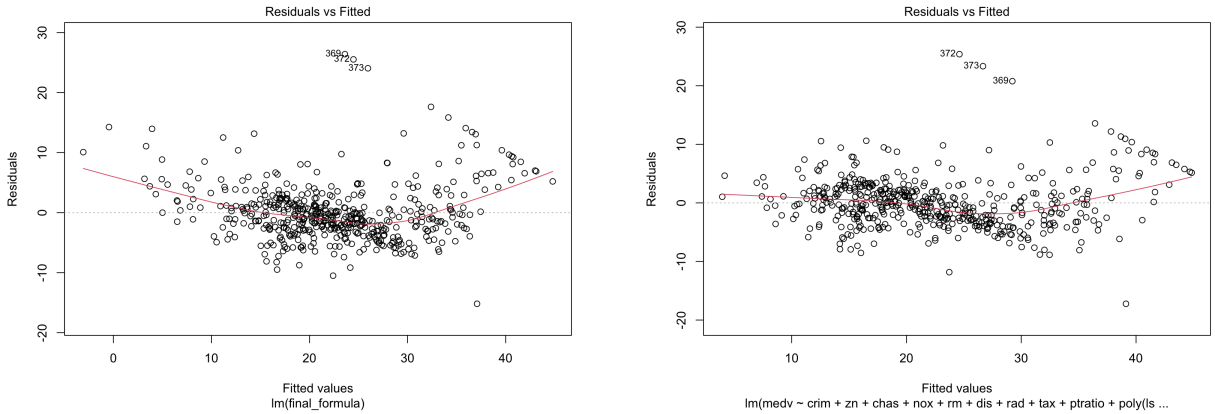

Figure 2: Diagnostics before/after transformation on lstat

The presence of multicollinearity was checked - there were a few scores larger than 4 that may warrant further investigation. But nothing was larger than our cutoff of 10, and since this model is built for prediction (not inference), this is not a cause of concern. My final model is as follows:

Predictor	least-square estimate
(Intercept)	32.16919
crim	−0.1612964
zn	0.02122889
chas	2.722435
nox	−14.25721
rm	3.112007
dis	−1.376345
rad	0.2575762
tax	−0.009964112
ptratio	−0.7749002
lstat	−103.0910
$lstat^2$	49.45799

Note that this model was refit on the full dataset in order to get better least-square estimates for the predictors. Of course, the bias and variance cannot be calculated for this model. As we've seen in simulations, this requires knowing the true model. If we knew the true model, then none of this estimation and model fitting would have been necessary in the first place.

Problem 2: Follow-up to in-class activity

Let's recap what we learned from our in-class activity from 3/6/24 (see the post on Ed Discussion if you need a refresher).

(a) Clearly the models in Part 2 and Part 3 are not equivalent. Explain again in plain language (using plots if you deem it helpful) why these two approaches will not give the same results.

Solution: *The model in Part 2 assumes that* `bmi` *and* `age` *have the same effect on health care charges (Y) for both genders. In Part 3, we allow* `bmi` *and* `age` *to have different effects on Y for male and female separately.*

(b) Load in the Insurance dataset again. Show how you can modify the model in Part 2 so that it is exactly equivalent to the models in Part 3. Present your code and results to show that the fitted models for males only and females only are exactly the same in Part 2 and Part 3 (after making your modification). *Hint: use interaction terms.*

Solution: *The modified model (adding interaction terms involving gender) is*

$$\hat{Y} = -4515.22 + 246.92\,age + 241.31\,bmi - 3497.57\,gender + 168.55\,(bmi \cdot gender) - 8.29\,(age \cdot gender).$$

If gender is coded so that the reference group is female, then for females,

$$\hat{Y} = -4515.22 + 246.92\,age + 241.32\,bmi.$$

For males,

$$\begin{aligned}\hat{Y} &= (-4515.22 - 3497.57) + (246.92 - 8.29)\,age + (241.31 + 168.55)\,bmi \\ &= -8012.79 + 238.63\,age + 409.87\,bmi.\end{aligned}$$

By adding the interaction terms, the modified model for Part 2 becomes exactly equivalent to the model in Part 3:

Part 3 model for male only: $\hat{Y} = -8012.79 + 238.63 \cdot age + 409.87 \cdot bmi$

Part 3 model for female only: $\hat{Y} = -4515.22 + 246.92 \cdot age + 241.32 \cdot bmi$

(c) Other than domain knowledge, how might you determine using (in a data-driven way) that an interaction term is needed in the model? Justify and then implement your approach here on the Insurance dataset.

Solution: *There are a variety of ways students could have gone about this. The simplest approach would be to compare the test MSE of a model with/without the interaction term. They could have also checked if the model with the interaction terms was significantly better (using an F-test). The test MSE of my model without the interaction term was smaller than the model with the interaction term. The F-test was also non-significant at* $\alpha = 0.05$. *Therefore, I would conclude that the interaction term is not needed in the model. See my code below:*

```
index = sample(1:nrow(insurance),nrow(insurance)/2)
train = insurance[index,]
test = insurance[-index,]
model1 = lm(charges~bmi+age+gender,data=train)
```

```
model2 = lm(charges~bmi+age+gender+gender*age+gender*bmi,data=train)
yhat1 = predict(object=model1,newdata=test)
testMSE1 = mean((test$charges-yhat1)^2)
yhat2 = predict(object=model2,newdata=test)
testMSE2 = mean((test$charges-yhat2)^2)
testMSE1
testMSE2
anova(model1, model2)
```

Problem 3: Predictions in the presence of multicollinearity *(practice)*

(a) Is multicollinearity a problem for making accurate predictions? Make an educated guess based on what we have learned in class.

Solution: *Solutions will vary but the short answer is no. As long as students make a reasonable guess here (regardless of whether or not it is correct), you can give students full credit.*

(b) Let's carry out a simulation study to answer this. We will simulate data with and without multicollinearity. This is our true model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i$$

where $i = 1, \ldots, 100$, $\epsilon \sim N(\mu = 0, \sigma^2 = 4)$, $\beta_0 = 3$, $\beta_1 = 2$, and $\beta_2 = 4$.

To generate your predictors with multicollinearity use the following code:

```
set.seed(42)
x1 = runif(100)
x2 = 0.8*x1 + rnorm(100,0,0.1)
```

Using these predictors, generate your Y values in R. Check the correlation between `x1` and `x2` using the `cor()` function. Report that value here.

Solution: *The correlation between X_1 and X_2 is 0.9404249.*

(c) Split your data into a training set and test set. Train your model on the training set.

```
lm(Y ~ x1+x2, data = train)
```

Report the test MSE for this model.

Solution: *Solutions will vary.*

(d) Repeat this process 2,500 times. Each time you'll need to generate a random set of Y's, fit a model on your training set, and then obtain the test MSE for that model. We do not need to generate new predictor values (again, think about why). Remember to store the test MSE for each iteration and do not set seed. What is the mean test MSE in this setting when the predictors are highly correlated? Plot a histogram of your 2,500 test MSEs and comment on what you observe.

Solution: *Solutions will vary. The mean test MSE I obtained was 4.28. The distribution is roughly bell-shaped.*

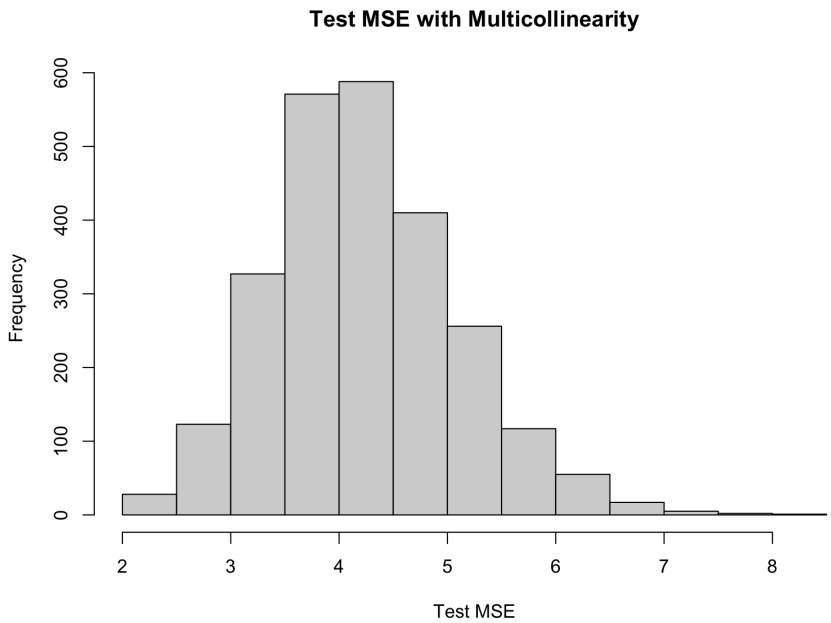

Figure 3: Test MSE with Multicollinearity

(e) Now generate predictors without multicollinearity using the following code:

```
set.seed(24)
x1 = runif(100)
x2 = rnorm(100,0,1)
```

Using these predictors, generate your Y values in R. Check the correlation between `x1` and `x2`. Report that value here.

Solution: *The correlation between X_1 and X_2 is 0.03316596.*

(f) Again run 2,500 simulations to obtain the test MSE of our model when the predictors are not correlated. What is the mean test MSE in this setting when the predictors are not correlated? Plot a histogram of your 2,500 test MSEs and comment on what you see.

Solution: *The test MSE I obtained was 4.26. The overall shape and distribution of the test MSEs look very similar to the setting when X_1 and X_2 were not correlated.*

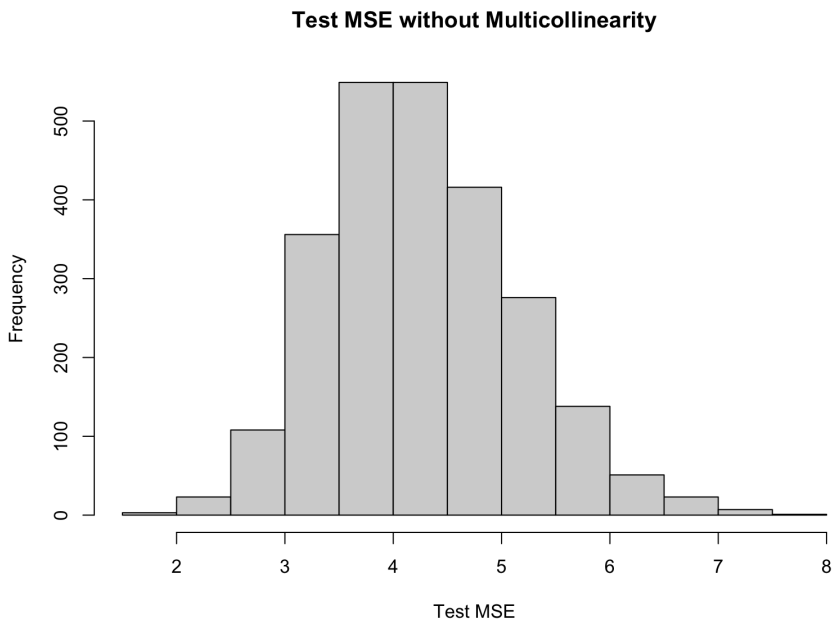

Figure 4: Test MSE without Multicollinearity

(g) Based on our simulation study, is multicollinearity a problem for making accurate predictions? Comment on your findings.

Solution: *Students should have observed that multicollinearity is not a problem in making predictions. Multicollinearity only affects the standard errors of our least square estimates but it does not affect our ability to obtain $\hat{Y}$. We can see that the mean test MSEs are close and both distributions of test MSEs look very similar.*

End of assignment.